STIVEN CATALYST

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MANAGEMENT REVIEW

Management without theatre.



OPERATIONS

POKA-YOKE: THE MISTAKE NOT ALLOWED TO HAPPEN

Two springs on a dish, 112 devices and what they cost, three kinds of inspection, why warnings go unheard, shape, count and sequence, and a card for one poka-yoke.

112

poka-yoke devices from Japanese plants in Shigeo Shingo's book, each one with its cost.

SHINGO, 1986

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Telling people to be more careful works for a while. The other way is to build the check into the work, so that a slip is caught before it becomes a defect, or cannot be made at all.

In 1961 Shigeo Shingo watched a worker leave a spring out of a switch and solved it with a small dish. His book of 1986 describes 112 such devices; more than half cost $100 or less, in 1986 dollars. Shingo separates inspections that find defects, reduce them and eliminate them, and he prefers a device that stops the work to one that only warns. Hospital alarms show why: in 98 events reported to the Joint Commission, the factor named most often among those listed was an alarm turned off.

Stiven Janaqi EDITOR · STIVEN CATALYST

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HOW TO READ THIS ISSUE

Figure
Every figure has its source and year at the foot of its page.

Our reading
Where the editor interprets rather than the research, it says so.

Practice
The steps and the card are proposals to try, not research results.

COVER STORY

Two springs on a dish

In 1961 Shigeo Shingo visited Yamada Electric in Nagoya. The plant made a simple switch with two buttons for its parent company, Matsushita Electric, with a small spring under each button. Now and then a worker left a spring out, and each time Yamada had to send an inspector to Kyushu to check every switch delivered.

Telling workers to be careful helped for a while, then it happened again. On the floor Shingo saw a worker skip a spring, and asked the department chief:

“What does it mean for a human being to ‘forget’ something?”

The answer was a dish. At the start, the worker took two springs from a box of hundreds and put them on it; a spring left on the dish showed that one had been left out. No more springs went missing, Shingo writes. Around 1963 he renamed such devices: “foolproofing” made a worker cry, so he chose poka-yoke, proofing against inadvertent mistakes.

OUR READING

The dish leaves memory out of it: it shows the forgetting before the switch leaves the bench.

Source: Shigeo Shingo, translated by Andrew P. Dillon, Productivity Press, 1986 (via Grout, 2007, and a scan of the book)

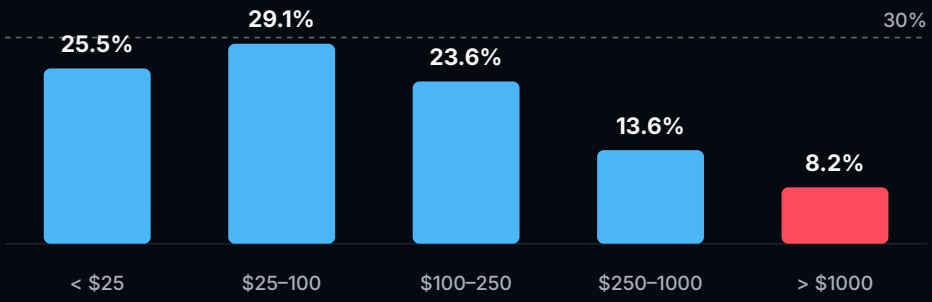
Shingo's own account, in Zero Quality Control (Japanese 1985, English 1986); we found no independent record. It is the first device he describes; foolproofing for safety came earlier.

THE NUMBERS

112 devices, most of them cheap

Shingo's book ends with 112 poka-yoke examples from Japanese plants, each with its cost. In 2007 John Grout sorted the costs in a report for the US Agency for Healthcare Research and Quality:

SHINGO'S EXAMPLES BY COST, IN US DOLLARS OF 1986



83 of 112
stop the work when an error occurs; the other 29 only warn

3.5% → 0.01%
defect rate at Arakawa Auto Body within two years, mainly through poka-yoke

OUR READING

When a device costs less than the hour spent arguing about it, what decides is whether anyone goes looking for the error.

Sources: John Grout, Agency for Healthcare Research and Quality, 2007; Shigeo Shingo, translated by Andrew P. Dillon, Productivity Press, 1986 (via Grout, 2007, and a scan of the book)

Shares of the examples, as tabulated by Grout; the median is about $100. The count of 83 is the sum of Shingo's own grouping. Arakawa is his account, with no year given.

THE MODEL

Find, reduce, eliminate

Shingo sorts inspections by what they do to defects. Errors, he argues, cannot be removed from human work. But an error becomes a defect only if nothing happens at the stage where it is made.

01

Judgment inspection

sorts finished products into good and bad: it protects the customer, but does not lower the defect rate

02

Informative inspection

uses the defects found to correct the process: statistical control, checks at the next step, self-checks

03

Source inspection

checks the conditions that give rise to defects and acts at the error, before it becomes a defect

At Matsushita's Moriguchi television plant in 1963, statistical control had cut a 15% defect rate to 6.5%, where it stuck. Three months after checks at the next step began, it was 0.65%. Around 1967 Shingo arrived at source inspection. His Zero QC adds: check every item, not a sample, and act at once.

OUR READING

Every defect found at final inspection was made at some earlier step. The model asks which step, and what there let the slip through.

Source: Shigeo Shingo, translated by Andrew P. Dillon, Productivity Press, 1986 (via Grout, 2007, and a scan of the book)

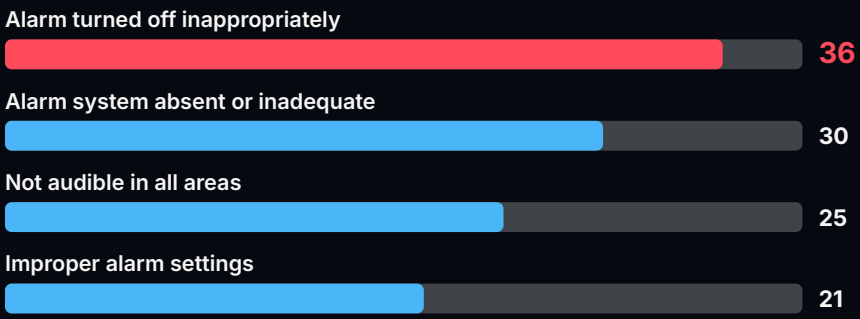
The three inspections and the Moriguchi rates are Shingo's account (1986). The conclusions drawn here are the editors'.

WHAT THE RESEARCH SAYS

When the warning goes unheard

Shingo ranks devices that stop the work above those that only light a lamp or sound a buzzer: a warning works only if someone notices. In 2013 the Joint Commission counted 98 alarm-related events reported to it from 2009 to mid-2012; 80 ended in death.

CONTRIBUTING FACTORS IN THE 98 EVENTS, NUMBER OF EVENTS



The alert estimates that 85 to 99% of alarm signals need no clinical action; a 2013 review puts false alarms at 72 to 99%. In 2026 the Institute for Safe Medication Practices ranked forcing functions highest, warnings and checklists in the middle, and reminders to “be more careful” among the lowest.

OUR READING

An alarm that sounds all day becomes noise. Before adding one, ask whether the step could be stopped.

Sources: Shigeo Shingo, translated by Andrew P. Dillon, Productivity Press, 1986 (via Grout, 2007, and a scan of the book); The Joint Commission, 2013; Sue Sendelbach & Marjorie Funk, AACN Advanced Critical Care, 2013 (via PubMed abstract); Institute for Safe Medication Practices, ISMP Medication Safety Alert! Acute Care 31(1), 2026

Reporting is voluntary and covers a small share of events; one event can have several factors. The counts show kinds of failure and say nothing about how often each occurs.

HOW IT IS MEASURED

Shape, count, sequence

Shingo names three ways for a device to detect an error. Then it either stops the work, the stronger response, or warns with a light or a sound.

01	**Contact**	checks shape or size: the part fits the guide or touches the sensor, or it does not
02	**Fixed value**	checks that a step was repeated the set number of times, like six nuts welded on one part
03	**Motion step**	checks that the standard motions were done, in their order

HYPOTHETICAL EXAMPLE, A LOADING RAMP FOR HOME DELIVERIES

Fixed value	bags scanned against the order; the trolley is not released until the count matches
Test	once a shift, a trolley with one bag missing: does the block hold?

A device nobody tests can fail unseen. The ramp is a made-up case.

Sources: Shigeo Shingo, translated by Andrew P. Dillon, Productivity Press, 1986 (via Grout, 2007, and a scan of the book); Lean Enterprise Institute, 2014; John Grout, Agency for Healthcare Research and Quality, 2007

The three methods and the six nuts are Shingo's. The Lean Enterprise Institute calls the stopping kind shutdown devices, the most powerful. Grout's advice: mistake-proof the mistake-proofing.

More in the essay Where part of the order goes missing

TOOL OF THE ISSUE

The poka-yoke card

One card for one recurring error. Start where the error is made, not where the defect is found, and ask what in the work makes the slip possible before you ask for more attention. The Incomplete Control tool shows where orders go out short.

01 The error what is left out, swapped or done backwards; how often

02 Where it starts the step where it is made

03 Inspection today judgment, informative or at the source

04 Method contact, fixed value or motion step

05 Stop or warn if it only warns: who reacts, and how fast

06 Test and owner how often the device is tested, by whom, what happens when it fails

Sources: Shigeo Shingo, translated by Andrew P. Dillon, Productivity Press, 1986 (via Grout, 2007, and a scan of the book); John Grout, Agency for Healthcare Research and Quality, 2007

The editors propose the card, after Shingo (1986) and Grout (2007).

Open the tool: Incomplete Control stivencatalyst.com/tools/incomplete-control/

SOURCES AND METHOD

Every figure has a source.

The figures in this issue come from the sources below. The year shows how recent each one is.

Zero Quality Control: Source Inspection and the Poka-yoke System
Shigeo Shingo, translated by Andrew P. Dillon, Productivity Press, 1986 · via Grout, 2007, and a scan of the book
https://www.routledge.com/Zero-Quality-Control-Source-Inspection-and-the-Poka-yoke-System/Shingo/p/book/9780915299072

Mistake-Proofing the Design of Health Care Processes (AHRQ Publication No. 07-0020)
John Grout, Agency for Healthcare Research and Quality, 2007
https://www.govinfo.gov/content/pkg/GOVPUB-HE20_6500-PURL-gpo3107/pdf/GOVPUB-HE20_6500-PURL-gpo3107.pdf

Lean Lexicon: Poka-yoke
Lean Enterprise Institute, 2014
https://www.lean.org/lexicon-terms/poka-yoke/

Sentinel Event Alert, Issue 50: Medical device alarm safety in hospitals
The Joint Commission, 2013
https://digitalassets.jointcommission.org/api/public/content/f65e5c9df2b94000a99445e0a7877007

Alarm fatigue: a patient safety concern
Sue Sendelbach & Marjorie Funk, AACN Advanced Critical Care, 2013 · via PubMed abstract
https://doi.org/10.1097/NCI.0b013e3182a903f9

Implement High-Leverage and Layered Risk-Reduction Strategies Using ISMP's Hierarchy of Effectiveness
Institute for Safe Medication Practices, ISMP Medication Safety Alert! Acute Care 31(1), 2026
https://home.ecri.org/blogs/ismp-alerts-and-articles-library/implement-high-leverage-and-layered-risk-reduction-strategies-using-ismp-s-hierarchy-of-effectiveness

EDITORIAL METHOD

Each figure was checked for its year, its publisher and what it measures, in the full text where it could be opened. A source read only in a summary, an abstract or excerpts is marked “via”. A figure not confirmed in what was read is marked so on its page, or left out. The editor's interpretation is marked “Our reading”.



MANAGEMENT REVIEW

Management without theatre.

Every issue, one management question, checked against the best research.

ALL ISSUES

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Edited by Stiven Janaqi, who answers for it; “the editors” in this publication means him. Its texts, translations, layout and the checking of figures against their sources are done in working sessions with Claude, an AI model by Anthropic.
stivencatalyst.com/how-this-is-made.html